Arista Access Node ELEC-ACC-01 Routing Protocol Flap Resolution SOP

Node: ELEC-ACC-01 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_5617

Description

This Standard Operating Procedure (SOP) outlines the steps required to troubleshoot and resolve a Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) failure on the Arista Access Node ELEC-ACC-01 located at the Electronic City Hub. The failure domain is characterized by the alarm code ALARM_5617 and a severity of Minor. The underlying cause of this failure domain is attributed to the dynamic nature of routing protocols, which can lead to frequent route flaps due to factors such as BGP route churn, OSPF adjacency loss, and ISIS LSP flooding. These events can be triggered by various factors, including network topology changes, routing configuration updates, or hardware issues. As the Access Node ELEC-ACC-01 is equipped with Arista's proprietary EOS (Extensible Operating System) and features a high-performance hardware platform, the likelihood of routing protocol flaps is further exacerbated by the increased network traffic and complexity.

Pre-requisite Checks

- check: Verify that the Electronic City Hub is accessible and that the remote hands are dispatched to the site., description: Ensure that the site is accessible and that the necessary personnel are present to facilitate the troubleshooting and resolution process.

- check: Review the redundant link status to ensure that it is functioning correctly., description: Verify that the redundant link is operational and can take over in the event of a failure to prevent network outages.

Potential Root Causes

Cause	Probability
BGP Route Churn	High
OSPF Adjacency Loss	Medium
ISIS LSP Flooding	Low

Diagnostic Commands & Expected Outputs

```
Command: show ip bgp summary, Expected Output: BGP table version is 1, main routing table version is 1 BGP router identifier is 10.10.10.10, local AS number 64512 Neighbor V AS MsgRcvd MsgSent TblVer InQ OutQ Up/Down State/PfxRcd, Description: This command displays a summary of the BGP neighbor connections and can help identify any issues related to BGP route churn.

Command: show ospf neighbor, Expected Output: OSPF Neighbor Table Neighbor ID Pri State Dead Time Address Interface 10.10.10.10 1 FULL/
```

```
- 00:00:32 10.10.10.10 eth1 , Description: This command displays a list of OSPF neighbors and can help identify any issues related to OSPF adjacency loss.

Command: show isis database, Expected Output: ISIS Database Level 1 Database Sequence Number: 1 IS-IS Router: ELEC-ACC-01 ISIS Metric: 10 ISIS LSP ID: 10.10.10.10 ISIS Sequence Number: 1 , Description: This command displays a list of ISIS LSPs and can help identify any issues related to ISIS LSP flooding.
```

Resolution Steps

Step 1: Step: Clear BGP Neighbor Connections, Command: clear ip bgp *, Description: This command clears all BGP neighbor connections and can help resolve issues related to BGP route churn.

Step 2: Step: Re-establish OSPF Adjacency, Command: clear ospf neighbor *, Description: This command re-establishes OSPF adjacency and can help resolve issues related to OSPF adjacency loss.

Step 3: Step: Re-originate ISIS LSP, Command: clear isis database, Description: This command re-originates the ISIS LSP and can help resolve issues related to ISIS LSP flooding.

Post-Check Verification

```
Command: show ip bgp summary, Expected Output: BGP table version is 1, main routing table version is 1 BGP router identifier is 10.10.10.10, local AS number 64512 Neighbor V AS MsgRcvd MsgSent TblVer InQ OutQ Up/Down State/PfxRcd, Description: This command verifies that the BGP neighbor connections have been resolved.

Command: show ospf neighbor, Expected Output: OSPF Neighbor Table Neighbor ID Pri State Dead Time Address Interface 10.10.10.10 1 FULL/ - 00:00:32 10.10.10.10 eth1 , Description: This command verifies that the OSPF adjacency has been re-established.

Command: show isis database, Expected Output: ISIS Database Level 1 Database Sequence Number: 1 IS-IS Router: ELEC-ACC-01 ISIS Metric: 10 ISIS LSP ID: 10.10.10.10 ISIS Sequence Number: 1 , Description: This command verifies that the ISIS LSP has been re-originated.
```

Roll-back Procedures

- Step: Restore BGP Neighbor Connections, Command: show ip bgp summary, Description: This command restores the BGP neighbor connections to their previous state.

- Step: Re-establish OSPF Adjacency, Command: clear ospf neighbor *, Description: This command re-establishes OSPF adjacency to its previous state.

- Step: Re-originate ISIS LSP, Command: clear isis database, Description: This command re-originates the ISIS LSP to its previous state.